AAbAAC: annotation guide

What we want to do

We aim to train a language model to automatically:

1. detect entities of the following types in English-language texts (PubMed abstracts): Autoimmune disease, their associated clinical sign, autoantibodies, and their targets and potential locations;
2. extract relations of specific types between these entities.

To do that, we propose to manually annotate a corpus where each entity and relation is clearly typed and precisely delimited, to train the model.

We need annotators with scientific and/or medical knowledge to manually annotate this corpus that we will use to train the model.

This document details the rules the annotators should follow while annotating the texts. These rules may be modified, if annotators suggest changes that could make their work better or easier. The document you are currently reading was last edited on 20-04-2026.

Types of objects

Entities

Type of entity	Definition
Disease	A **disease** is an abnormal condition that adversely affects the structure or function of all or part of an organism and is not immediately due to any external injury. Diseases are often known to be **medical conditions** that are associated with specific signs and symptoms. A disease may be caused by external factors such as pathogens or by internal dysfunctions. For example, internal dysfunctions of the immune system can produce a variety of different diseases, including various forms of immunodeficiency, hypersensitivity, allergies, and autoimmune disorders.
Autoantibody	An **autoantibody** is an antibody (a type of protein) produced by the immune system of an organism and is directed against molecules of the same organism. An antibody will not be

	considered an autoantibody if it comes from outside the organism (ex: isoantibody, autoantibody used as treatment). Do not annotate an antibody isotype alone (eg. “IgG”), but include the isotype if it comes with an autoantibody name.
Clinical signs and symptoms (shall be referred to as: “**Clinical_sign**”, for convenience)	**Signs and symptoms** are the observed or detectable signs, and experienced symptoms of an illness, injury, or condition. **Signs** are objective and observable; symptoms are a person's reported subjective experiences.[1] A sign for example may be a temperature higher or lower than normal, raised or lowered blood pressure or an abnormality showing on a medical scan. The presence of an autoantibody, revealed by a test, is a biological sign. A **symptom** is something out of the ordinary that is experienced by an individual such as feeling feverish, a headache or pain. In some cases, signs and symptoms can also be diseases.
Autoantibody target (shall be referred to as: “**Target**”, for convenience)	Organ, tissue, cell, protein, or by extension an epitope damaged or recognized by an autoantibody; when it is mentioned as a target of an autoantibody (it should not be annotated in a context when it is not mentioned as an autoantibody target).
Autoantibody location (shall be referred to as: “**Location**”, for convenience)	The location of an Autoantibody in a tissue, organ or structure of the body (e.g. cerebrospinal fluid, blood); where it can be found. A location is not an autoantibody target, nor the location of a Target.

Relations

Type of relation	Definition	Examples Predicate(subject, object)
isAssociatedWith **domain**: Disease, Autoantibody, Clinical_sign **range**: Autoantibody, Clinical_sign, Disease	Relation to use when the text explicitly indicates a link of any type between 2 entities, if a more specific relation cannot be used. (“Entity1” isAssociatedWith “Entity2”)	anti-Ma2-associated paraneoplastic syndrome isAssociatedWith(“anti-Ma2”, “paraneoplastic syndrome”)

	Do not link 2 autoantibodies. One of the two entities must be an Autoantibody, except in the case of a link between a Disease and a Clinical_sign if the Clinical_sign mentions an Autoantibody **(symmetrical relation)**	
isNotAssociatedWith **domain:** Disease, Autoantibody, Clinical_sign **range:** Autoantibody, Clinical_sign, Disease	Relation to use when the text denies a link of any type between 2 entities or if there is a suspected negative association between 2 entities. isNotAssociatedWith ("Entity1", "Entity2") **(symmetrical relation)**	Thyroid cancer was not associated with elevated levels of thyroid peroxidase antibodies isNotAssociated("thyroid cancer", "elevated levels of thyroid peroxidase antibodies")
mayBeAssociatedWith **domain:** Disease, Autoantibody, Clinical_sign **range**: Autoantibody, Clinical_sign, Disease	Relation to use when a relation between 2 entities is clearly mentioned as uncertain. Otherwise, by default, use isAssociatedWith. mayBeAssociatedWith("Entity1", "Entity2") **(symmetrical relation)**	anti-Saccharomyces cerevisiae antibody (ASCA) may be associated with pouchitis after ileal pouch-anal anastomosis mayBeAssociatedWith("anti-Saccharomyces cerevisiae antibody", "pouchitis")
isCausedBy **domain**: Disease, Clinical_sign **range**: Autoantibody, Disease, Clinical_sign	Relation to use when the text indicates an entity is caused by another. isCausedBy("Consequence", "Cause")	anti-Ma2 is a well-characterized onconeural antibody and one of the causes of PCD isCausedBy("PCD", "anti-Ma2")

isTargetOf **domain**: Target **range**: Autoantibody	Relation to use when the text indicates a target is recognized by an autoantibody: isTargetOf (“Target”, “Autoantibody”) There is no need to manually annotate that relation when the target is included inside the antibody name, except when it is not immediately clear (example: multiple targets as in “anti-liver/kidney antibody”.	antibodies against human thyroid peroxidase isTargetOf(“human thyroid peroxidase”, “antibodies against human thyroid peroxidase”)
isRefTo **domain**: any entity **range**: any same type	Relation to use when a previously mentioned entity is mentioned again, or referred to in an indirect way in the same sentence, but is not directly specified as a synonym. Example: “Entity1 is linked to Entity2, and [...] it is also linked to Entity3.” => isRefTo(“It”, “Entity1”) (from 2ndary part to main part)	Systemic lupus erythematosus, [...] this disease is… isRefTo(“this disease”, “systemic lupus erythematosus”) (in the context of this example)
discontiguousEntity **domain**: any type **range**: any same type	Relation to use when an entity is composed of several discontiguous fragments (from 2ndary part to main part): discontiguousEntity(“Entity1_secondary_part”, “Entity1_main_part”)	lung and breast cancer discontiguousEntity(“lung”, “cancer”)
isSynonymOf **domain**: any type **range**: any same type	Relation to use when the text explicitly indicates 2 entities are synonymous. This includes abbreviations :	islet cell antibody (ICA) isSynonymOf(“ICA”, “islet cell antibody”)

	isSynonymOf("D1", "disease1"). The entity mentioned first in the text should be the subject of the relation. **(symmetrical relation)**	
isLocatedIn **domain**: Autoantibody, Target **range**: Location, Target	Relation to use in order to specify where an Autoantibody is located or where a Target is located (an Autoantibody can only be located in a Location; and a Target can only be located in a larger Target). isLocatedIn("AAb1", "Location1") isLocatedIn("Target2", "Target1")	The major antigens recognized by antihistone antibody are histones H1, H2A and H2B, which are located on the outer side of the nucleosome
		isLocatedIn("histone H1", "outer side of the nucleosome")
		Anti-Ri antibodies were positive in serum
		isLocatedIn("Anti-Ri antibodies", "serum")
isSubClassOf **domain**: any type **range**: any same type	Relation to use in order to describe a hierarchical relationship between 2 entities, where one of them is a more specific type of the other entity.	Oncocytic carcinoma is a rare malignant tumor
		isSubclassOf("Oncocytic carcinoma", "malignant tumor")

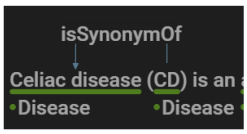
General rules

1. As a general rule, be as precise and exclusive as you can: if you can use a more specific relation, do it. If you can limit the annotation by excluding a few words from it without losing information, do it. Example: "a mild case of chronic hepatitis C" => this should be annotated as disease: "chronic hepatitis C", because the adjective "mild" is not part of

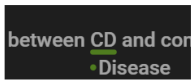
the disorder's name. However "chronic" is part of the name. There can be exceptions depending on the context.

2. Do not include determinants or articles in an annotation if they are the first word of that annotation. Also, do not include extra spaces, or punctuation.
3. A relation should not be annotated if the 2 entities linked by that relation are not in the same sentence.
4. Annotate only explicit relations. If a relation is implied but not written in the text (the annotator has to deduce it), then it should not be annotated.
5. When annotating autoantibodies Targets, be as specific as possible. If you can limit the annotation to part of a molecule, rather than the whole molecule, you should do that. E.g. : "outer side of the nucleosome" instead of "nucleosome"
6. When the name of a Disease or Autoantibody is first mentioned in a text along with a shortened form, do not include the shortened form in the main annotation (unless it is included inside the main entity). The abbreviation should have its own entity annotation, linked to the main name by the "isSynonymOf" relation (for its first occurrence). Then, following occurrences of the shortened form can be annotated like a regular entity. Example:

1st occurrence



Following occurrences



7. If an entity is enclosed between quotes, parentheses, brackets, or any other type of special characters, these special characters should not be included in the annotation.
8. Two entities can be related by "isRefTo", if both are two distinct mentions of the same entity, within the same sentence. "isRefTo" must not be used to refer to an entity outside the current sentence. If an entity is indicated to be a synonym of another "isSynonymOf" must be used.
9. "isLocatedIn" should not be used for sentences of type "epitope located on xxx". In that case, "xxx" should be annotated as a Target. If an indication more precise than "epitope" or more precise than a similar generic term is given, then that expression can be annotated as a Target located on the "xxx" Target.

Non-exhaustive list of ambiguous cases

1. **Disease with adjectives**: annotate only the disease without its adjectives, unless the name of the disease includes the adjectives. Example: in "juvenile dermatomyositis", "juvenile" should be included in the annotation because the full name of the disease is "juvenile dermatomyositis". If only "myositis" is included, it is not possible to discriminate with the disease "adult dermatomyositis". To know when a disease with symptoms is a separate disease, check if it has a code in SNOMED CT or Orphanet.
2. **Disease with symptoms/clinical signs**: annotate the disease and its symptoms separately, unless the disease with these specific symptoms is a different subtype of the disease. Example:

"eosinophilic granulomatosis with polyangiitis", shortened as "EGPA", is a disease of its own. Here "polyangiitis" should not be annotated separately. To know when a disease with symptoms is a separate disease, check if it has a code in SNOMED CT or Orphanet.

3. **Disease that can also be a symptom / symptom that can also be a disease**: if an expression could be annotated as a symptom or a disease, check in SNOMED or Orphanet if it is recorded as a finding or a disorder. If, in the context of the text, the annotator believes the expression is not of the type specified in SNOMED CT or Orphanet, then the expression should be annotated as both types.

4. **Abbreviation in the middle of an entity**: when an abbreviation is included in the longer expression it stands for, the abbreviation should not be annotated separately and linked using discontiguousEntity relation. Instead there should be only one entity annotation, including the abbreviation. If the abbreviation comes before or after the larger entity it stands for, then it should be annotated as a separate entity, and linked with refTo or isSynonymOf relations.
5. **Several entities mentioned together in an inseparable way**
   For example in "lung and breast cancer", 2 diseases are mentioned: "lung cancer" and "breast cancer", but it is not possible to annotate them separately. So they should be annotated that way: discontiguousEntity("lung","cancer"); and discontiguousEntity ("breast", "cancer"). Note that in this case, "lung" and "breast" are annotated as disease, because they go with "cancer". The direction of the discontiguousEntity relation should be: the secondary entity pointing towards the main entity.

6. **Adjectives included inside an entity**

   Annotate as a single entity, including the adjective

7. **"Anti-X antibody", "X antibody positivity" (nested entities)**
   In the case of nested entities, all of them should be annotated, but not the relations between them. E.g. : antinuclear antibody (ANA)-positive => antinuclear antibody (ANA)-positive, antinuclear antibody (ANA)-positive. The relations between these entities will be automatically annotated later.
8. **Clinical sign written in short and/or composite form**
   If several short forms are part of a composite short form, annotate each short form as Clinical_sign. If possible, annotate the 1st occurrence of each form with isSynonymOf their longer form. The composite form should also be annotated as a clinical sign, and if possible with isSynonymOf the longer composite form. For example, in one paper, "VT+/PM-" refers to "vascular thrombosis without pregnancy morbidity". This leads to the following annotations:
   - "VT+" -> Clinical_sign
   - "PM-" -> Clinical_sign
   - "VT+/PM-" -> Clinical_sign
   - "VT+" isSynonymOf "vascular thrombosis"
   - "PM-" isSynonymOf "no pregnancy morbidity"
   - "VT+/PM-" isSynonymOf "vascular thrombosis without pregnancy morbidity"
9. **Uncertain association**

When a link of any nature between 2 entities is suspected, it should only be annotated with "mayBeAssociatedWith" if that suspected association is clearly described as uncertain. Otherwise, it should be annotated with "isAssociatedWith".

10. **Text partially or entirely in non-English language, or non-latin script**
    Annotate only the English part. Ideally write down the reference of that text ro report it.
11. **Text unrelated to the topic of autoantibodies**
    Skip it and do not annotate. Ideally write down the reference of that text to report it.
12. **Biological tests / exams**
    Test results (positive, or negative) should be annotated as "Clinical_sign". If a test is mentioned without a result, it should not be annotated.